

$$1) \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_j \\ \vdots \\ x_p \end{bmatrix}$$

Let $\beta_0 \in \mathbb{R}$, $\beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \in \mathbb{R}^p$ and $f: \mathbb{R} \rightarrow \mathbb{R}$

Ridge Regression $\hat{f} = \underset{f \in \mathcal{Y}}{\operatorname{argmin}} \left[\sum_{i=1}^N (y_i - f(x_i))^2 + \lambda \sum_{j=1}^p \beta_j^2 \right]$

Function class, $\mathcal{Y}$:

$$f(x) = \beta_0 + \beta'x$$

Loss function:

$$L(y, f(x)) = (y - f(x))^2, \text{ the squared-error loss}$$

Penalty function:

$$\mathcal{J}(f) = \sum_{j=1}^p \beta_j^2 = \|\beta\|_2^2, \text{ the } L_2\text{-Norm}$$

Null space, $\mathcal{Y}_0$:

$$f(x) = \beta_0$$

Lasso Regression $\hat{f} = \underset{f \in \mathcal{Y}}{\operatorname{argmin}} \left[\sum_{i=1}^N (y_i - f(x_i))^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$

Function class, $\mathcal{Y}$:

$$f(x) = \beta_0 + \beta'x$$

Loss function:

$$L(y, f(x)) = (y - f(x))^2, \text{ the squared-error loss}$$

Penalty function:

$$\mathcal{J}(f) = \sum_{j=1}^p |\beta_j| = \|\beta\|_1, \text{ the } L_1\text{-Norm}$$

Null space, $\mathcal{Y}_0$:

$$f(x) = \beta_0$$

Smoothing Splines $\hat{f} = \underset{f \in \mathcal{Y}}{\operatorname{argmin}} \left[\sum_{i=1}^N (y_i - f(x_i))^2 + \lambda \int f''(t)^2 dt \right]$

Function class, $\mathcal{Y}$:

$\mathcal{Y} = \{f : \int f''(x)^2 dx < \infty\}$, i.e. all real functions that are twice differentiable with finite total curvature

Loss function:

$$L(y, f(x)) = (y - f(x))^2, \text{ the squared-error loss}$$

Penalty function:

$$\mathcal{J}(f) = \int f''(t)^2 dt$$

Null space, $\mathcal{Y}_0$:

$$f(x) = \beta_0 + \beta'x \quad (f''(x) = 0)$$

3) a. Least Squares Regression:

$$\hat{\beta}_{OLS} = (X'X)^{-1} X'y$$

where $X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N1} & x_{N2} & \dots & x_{Np} \end{bmatrix} \overset{\text{vector of 1's}}{=} [x_{(0)}, x_{(1)}, x_{(2)}, \dots, x_{(p)}]$

$$\Rightarrow \underline{r} = y - X \hat{\beta}_{OLS}$$

$$X' \underline{r} = X'(y - X \hat{\beta}_{OLS})$$

$$= X'y - X'X(X'X)^{-1}X'y$$

$$= 0$$

$$\Rightarrow \langle x_{(j)}, \underline{r} \rangle = 0 \text{ for all } j = 0, 1, \dots, p$$

Hence, the residual vector is orthogonal to every column vector of X .

Covariance: $\frac{1}{N} \sum_{i=1}^N (x_{ij} - \bar{x}_j)(\hat{r}_i - \bar{r})$

$$= \frac{1}{N} \sum_{i=1}^N x_{ij} \hat{r}_i \quad (\text{Since matrix } X \text{ is standardised every column mean is } 0 \text{ \& } \bar{r} = 0 \text{ since } \langle x_{(0)}, r \rangle = \sum_{i=1}^N r_i = 0)$$

$$= 0 \quad \text{for all } j=0,1,\dots,p \quad (\text{Since } \langle x_{(j)}, \hat{r} \rangle = 0 \text{ for all } j=0,1,\dots,p)$$

The covariance between the residual vector and each column vector of the predictor matrix X is 0.

Therefore, the correlation between the residual vector and each column vector of the predictor matrix X is 0.

Hence, we have shown that the columns of the predictor matrix X are uncorrelated with the residual vector as a result of the residual vector being orthogonal to every column vector in X and every column in X having a mean of 0.